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AI School Iteration 2.3

Logistic Regression

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01. What is Classification ?

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Classification = Predicting a discrete label (class) as output

Examples:

- Email Spam Filtering (Spam / Not Spam)
- Medical Diagnosis (Benign / Malignant Tumors)
- Sentiment Analysis - Classifying Social Media Posts (+ / - / neutral)
- Churn Prediction (customer will leave ("churn") or stay ("not churn"))

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02. Binary Classification

02. Binary Classification

Output: $y \in \{0,1\}$

Example:

- Will this student pass? yes/no

Goal:

Learn a function that maps inputs → class label

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02. Binary Classification as a Probability

$$P(y=1 \mid x)$$

Input: $x \in \mathbb{R}^n$
Output: $P(y=1 \mid x) \in [0,1]$

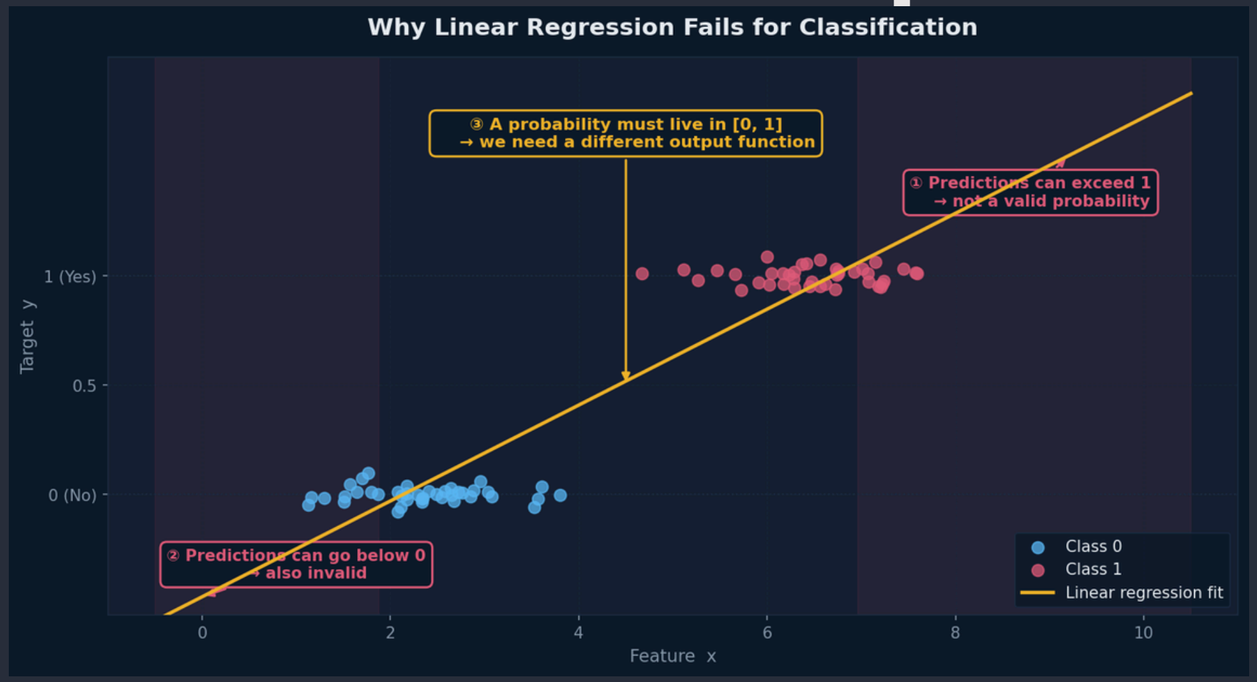
Thresholding (seuillage in fr):
$\hat{y} = 1$ if $P(y=1 \mid x) \geq 0.5$
$\hat{y} = 0$ otherwise

NB:

- Some models output labels directly (e.g. trees)
- Others output scores or probabilities

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02. Why Not Linear Regression?

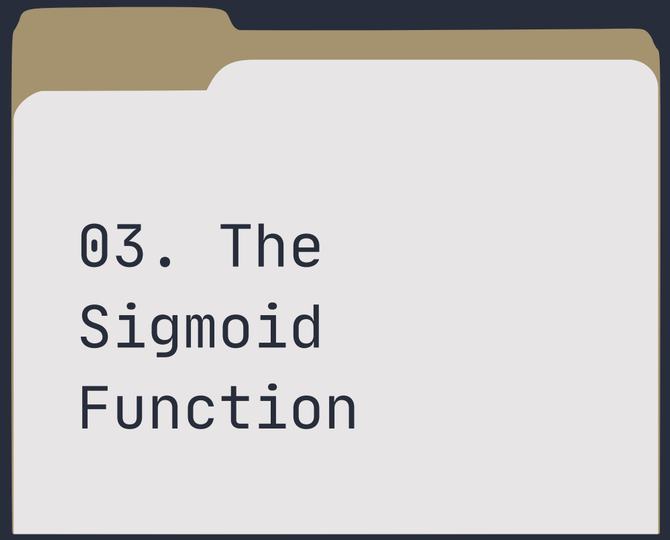


- Linear regression gives you outputs like `1.4`, `-0.2`, `0.87` , what do those mean?
- Nothing stops the line from predicting outside [0, 1]

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We need to restrict the output to [0, 1]



03. Sigmoid



The fix: Wrap the linear equation inside a sigmoid function that maps any real number → [0, 1].

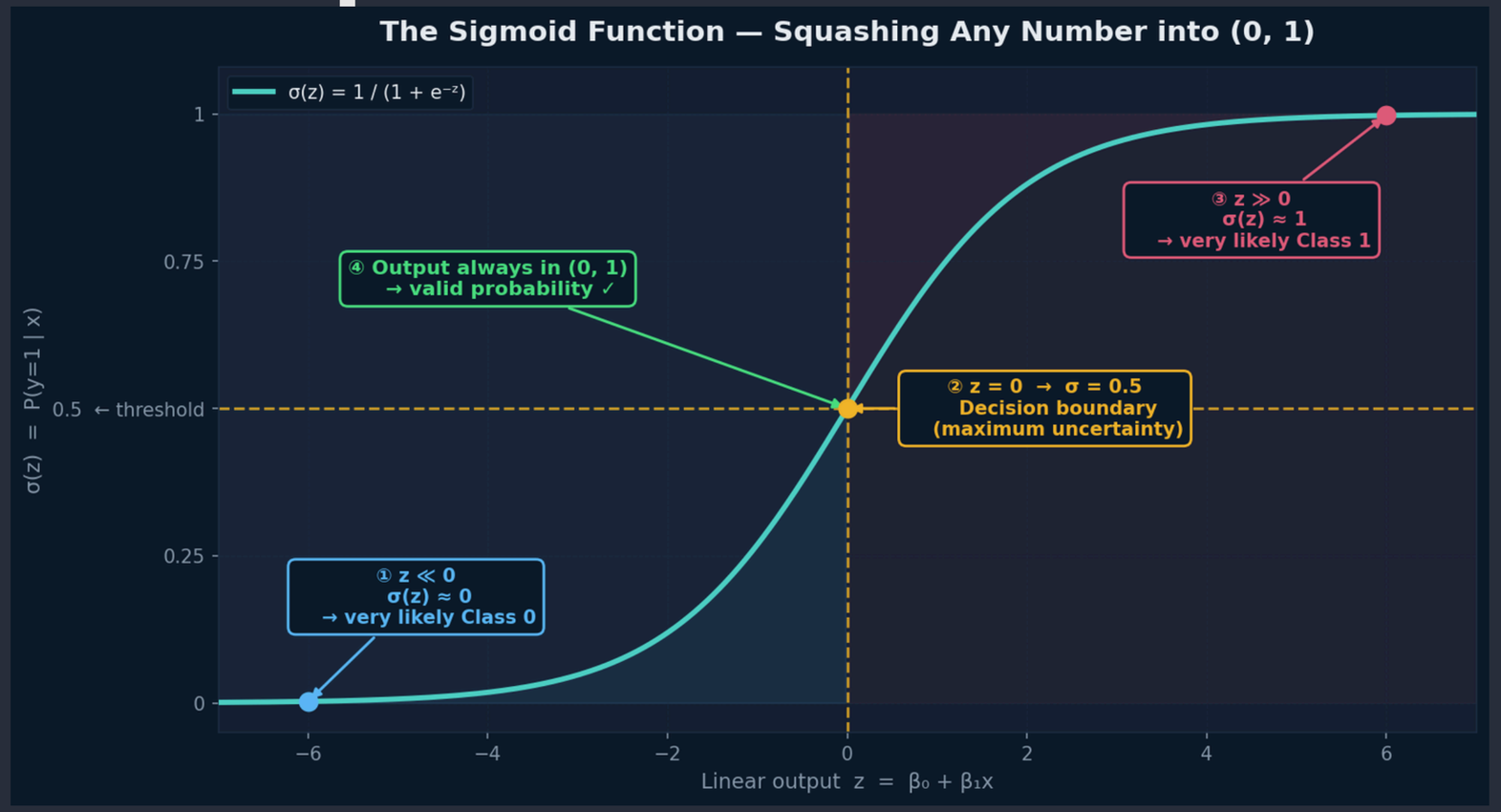
$$\sigma(z) = 1 / (1 + e^{-z}) \text{ where } z = \beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n$$

> So the full model is:
>
> $P(y=1 \mid x) = \sigma(h_\beta(X))$

X ---> h(x)
R^n ---> R

x ---> sigmoid(x)
R ---> [0, 1]

03. Sigmoid



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03. Logistic Regression: Forward Pass

$$X \in \mathbb{R}^n$$

Linear combination

$$z = \beta_0 + \beta^T X \quad \in \mathbb{R}$$

Sigmoid

$$P(y = 1 \mid X) = \sigma(z) = 1 / (1 + e^{-z}) \quad \in [0, 1]$$

Threshold (e.g. 0.5)

$$\hat{y} \in \{0, 1\}$$

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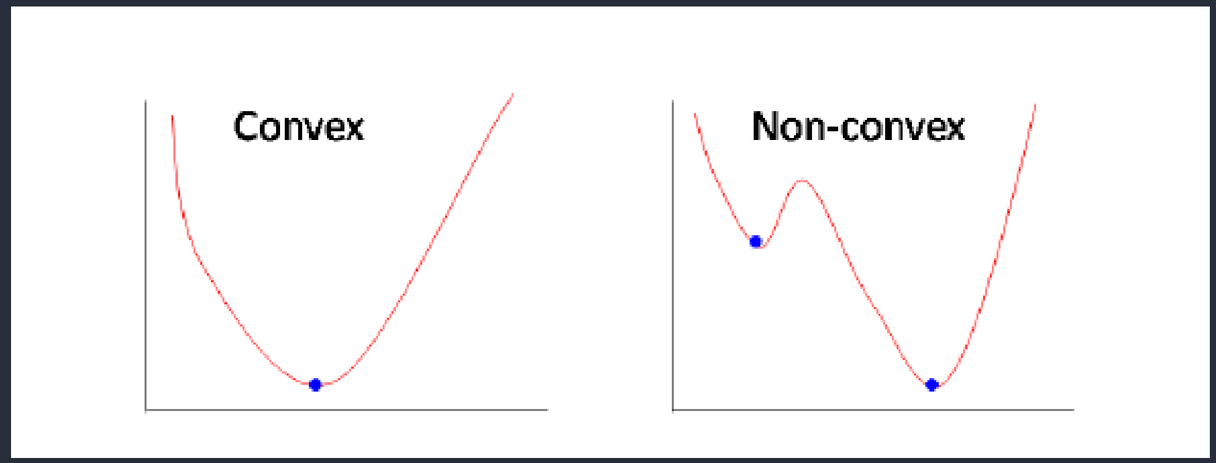
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04. Log Loss (Binary Cross-Entropy)

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Why not MSE?

- With sigmoid outputs, MSE creates a non-convex loss surface, gradient descent gets stuck in local minima



- MSE also doesn't penalize confident wrong predictions harshly enough

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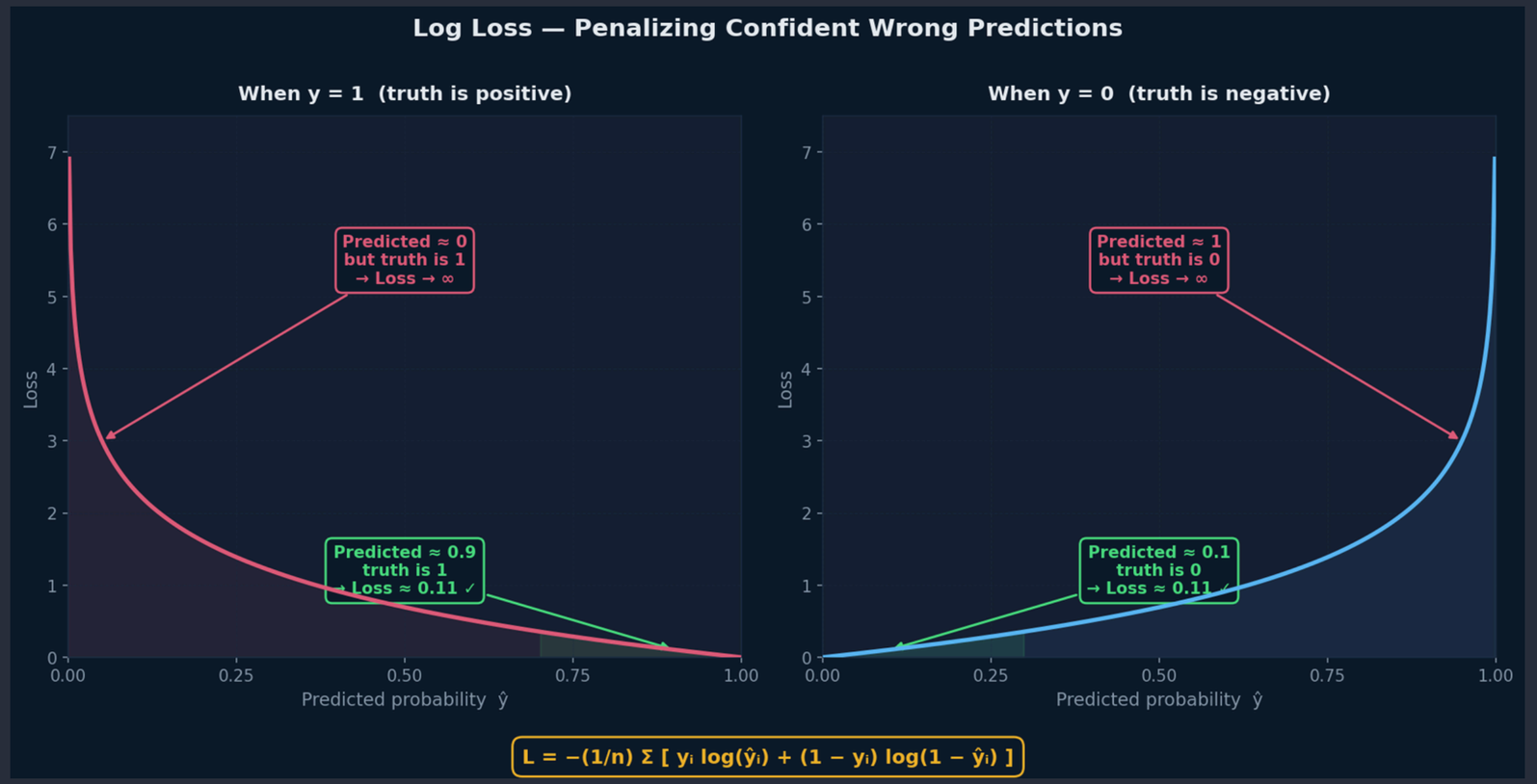
04. Log Loss (Binary Cross-Entropy)

$$L = -(1/n) \Sigma [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

Intuition:

Truth	Model says	Loss
y = 1	$\hat{y} \approx 1$	≈ 0 (good)
y = 1	$\hat{y} \approx 0$	→ ∞ (very wrong, heavily penalized)
y = 0	$\hat{y} \approx 0$	≈ 0 (good)
y = 0	$\hat{y} \approx 1$	→ ∞ (very wrong, heavily penalized)

04. Log Loss (Binary Cross-Entropy)



Key property: Log loss explodes when the model is confidently wrong. It rewards calibrated, honest predictions.

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05. Gradient Descent

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05. Gradient Descent

Good news: You already know this part.

The update rule of the backward pass is exactly the same:

$$w_i(t+1) = w_i(t) - \eta \cdot \partial L / \partial w_i$$

- The loss changed, but the algorithm does not change.
- Gradient descent on log loss is convex → guaranteed to find the global minimum

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Any Questions ?

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Thank you for your attention

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